THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL UNIT TEST SET 1

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. $(3x+2)(2x+5)$ equals: 1

(a) $6x^2+19x+10$ **(b)** $6x^2+10$ **(c)** $6x^2-19x+10$ **(d)** $6x^2+19x-10$

2. Two numbers are in the ratio 4 : 7 and their sum is 99. The larger number is: 1

(a) 72 **(b)** 36 **(c)** 49 **(d)** 63

3. An article is sold for ₹ 1260 at a profit of 5%. Its cost price (in ₹) is: 1

(a) 1197 **(b)** 1080 **(c)** 1200 **(d)** 1320

4. An isosceles triangle has equal sides 30 cm and base 36 cm. Its height (in cm) is: 1

(a) 24 **(b)** 23 **(c)** 25 **(d)** 12

5. If A:B = 3:4 and B:C = 8:9, then A:C is: 1

(a) 2:3 **(b)** 3:9 **(c)** 3:5 **(d)** 27:32

6. $(2x-3y)^2$ equals: 1

(a) $4x^2-12xy+9y^2$ **(b)** $4x^2-6xy+9y^2$ **(c)** $4x^2-9y^2$ **(d)** $4x^2+12xy+9y^2$

7. Two numbers are in the ratio 3 : 5. If 4 is added to each, the numbers become 28 and 44. The original smaller number is: 1

(a) 20 **(b)** 28 **(c)** 24 **(d)** 48

8. A watch is sold for ₹ 1530 at a loss of 15%. Its cost price (in ₹) is: 1

(a) 1620 **(b)** 1980 **(c)** 1800 **(d)** 1759.5

9. A rectangle has a diagonal of 30 cm and one side 18 cm. Its perimeter (in cm) is: 1

(a) 42 **(b)** 84 **(c)** 86 **(d)** 96

10. 12 workers working 8 hours a day finish a job in 15 days. 10 workers working 6 hours a day will finish it in (days): 1

(a) 24 **(b)** 18 **(c)** 180 **(d)** 26

11. The value of 63^2-37^2 is: 1

(a) 10000 **(b)** 5338 **(c)** 2600 **(d)** 676

12. A mixture of 60 L has milk and water in the ratio 7 : 3. The quantity of water is: 1

(a) 13 L **(b)** 42 L **(c)** 18 L **(d)** 23 L

13. After a discount of 15%, a pair of shoes is sold for ₹ 1530. The marked price (in ₹) is: 1

(a) 1620 **(b)** 1800 **(c)** 1980 **(d)** 1759.5

14. The side of a rhombus is 30 cm and one diagonal is 36 cm. The other diagonal (in cm) is: 1

(a) 24 **(b)** 48 **(c)** 49 **(d)** 50

15. The ratio of boys to girls in a school is 7:8. If there are 1350 students, the number of girls is: 1

(a) 720 **(b)** 675 **(c)** 630 **(d)** 1335

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Expand $(2x + 5y)^2$. 2

17. Two numbers are in the ratio 5 : 9 and their sum is 98. Find the numbers. 2

18. By selling an article for ₹ 1380, a trader gains 15%. Find the cost price. 2

19. An isosceles triangle has equal sides of 30 cm and base 36 cm. Find its height. 2

20. Two numbers are in the ratio 5:8 and their sum is 143. Find the numbers. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Expand $(3x + 2)(2x - 5)$. 3

22. A mixture of 40 L contains milk and water in the ratio 7 : 1. How much water must be added to make the ratio 7 : 3? 3

23. A shopkeeper allows successive discounts of 10% and 5% on a marked price of ₹ 4000. Find the selling price and the single equivalent discount percent. 3

24. A ladder 25 m long reaches a point 24 m up a wall. The foot is then pulled away so that it is 15 m from the wall. How far down does the top of the ladder slip? 3

OR

10 workers working 6 hours a day can finish a job in 24 days. In how many days can 12 workers working 8 hours a day finish it? 3

25. Find 197^2 using an algebraic identity. 3

OR

A : B = 1 : 2 and B : C = 6 : 5. If A + B + C = 150, find the three numbers. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A square plot has side $(3x + 2)$ m. A square pond of side $2x$ m is dug inside it. Write the area of the remaining land as an expression in x and find it when $x = 10$. 5

OR

12 men can complete a piece of work in 15 days. 5

(i) How many days will 18 men take?

(ii) How many men are needed to finish it in 9 days?

27. A trader sells a radio for ₹ 2100 and gains 5%. If he had sold it for ₹ 2400, what would be his gain or loss percent? 5

OR

A man walks 18 m east and then 24 m north. Find (i) the straight-line distance of his final position from the start, (ii) the distance saved by going straight, (iii) the total distance if he returns to the start by the straight path. 5

END OF QUESTION PAPER **Total: 50 Marks**

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